MUHAMMET MERT ERDOĞAN

Mechanical Engineer | Design, Analysis and Manufacturing
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Driving licence: B | Military service: Deferred | Willing to travel



ABOUT ME

I am interested not in one corner of mechanical engineering but in the whole path a product takes from idea to field. In my internships I worked across design, machining, assembly lines and on-site troubleshooting. I want to contribute to any team as an engineer who designs with manufacturing in mind and approaches manufacturing with design logic.

EDUCATION

Istanbul University-Cerrahpaşa *Sep 2026 - Present*
Institute of Graduate Studies, MSc in Mechanical Engineering (Thesis)

Istanbul University-Cerrahpaşa *Sep 2021 - Jul 2026*
Faculty of Engineering, BSc in Mechanical Engineering (30% English-medium)

EXPERIENCE

FESTO | Intern *Aug 2025 - Sep 2025*
- Traced mechanical, pneumatic, electrical and software faults in a robotic system via root cause analysis and recommissioned it.

BEKO | Manufacturing Intern *Jul 2025 - Aug 2025*
- Found loss sources at 5 critical assembly line stations through waste and root cause analysis and proposed 2 improvements.

MIDAS MAKİNA SANAYİ | Design Intern *Aug 2024 - Sep 2024*
- Revised 3D models of existing parts in SolidWorks and prepared manufacturing drawings to dimensioning and tolerancing standards.

YILMAZ REDÜKTÖR | Workshop Intern *Jul 2024*
- Observed CNC turning, milling, heat treatment and grinding operations and gear quality control on the production line.

PROJECTS

Beko | Capstone Project (NDA) *Oct 2025 - Jul 2026*
- Designed the conversion of a metal component to an injection-moulded engineering plastic part and validated it with topology optimisation and static FEA.

Mercedes-Benz Türk | Design Project (NDA) *Oct 2025 - Jul 2026*
- Developed an ergonomic luggage loading system and validated it with static and modal FEA.

İÜC Racing - Formula Student | Powertrain Team *Sep 2023 - Jul 2026*
- Designed the chain guard in SolidWorks and verified its strength with static analysis in ANSYS. Achieved a 15% weight reduction by revising the bracket design.

TEKNOFEST | UAV Landing Gear *Aug 2024 - Aug 2025*
- Modelled the landing gear parametrically in SolidWorks, optimised it with static FEA in ANSYS and produced it by FDM printing in PLA, ABS and ASA. The final weight came in 50% below the specification target.

Nuclear Reactor Pressure Vessel | Stress Analysis *Oct 2025 - Jan 2026*
- Calculated the membrane stresses with shell theory and compared them with an ANSYS model, showing that the head and flange discontinuities govern the design (safety factor 3.05 locally vs 11.2 in the shell). Extended the study to stored energy, TNT equivalent and blast safety distance, being prepared as an article.

Defect Prediction in Steel Production with Machine Learning *Feb 2025 - Jun 2025*
- Compared 7 classifiers in R and combined random forest, SVM and XGBoost by majority vote to predict the steel plate defect type with 91.3% accuracy.

SKILLS

Software: SolidWorks (CSWP), ANSYS Mechanical, AutoCAD, MS Excel, R, MATLAB/Simulink, Fusion 360, Siemens NX, CATIA
Expertise: Finite element analysis (static, modal), topology optimisation, solid modelling and parametric design, sheet metal design, design for injection moulding, technical drawing and tolerancing (GD&T), design for manufacturability (DFM), root cause analysis (RCA), lean manufacturing, machining processes
Languages: Turkish (native), English (C1)

CERTIFICATIONS AND TRAINING

- CSWP - Certified SOLIDWORKS Professional (Dassault Systèmes), Oct 2026
- UAV Design and Build Training - TEİ Aviation Engine Schools | McKinsey & Company Forward Program

MEMBERSHIPS AND LEADERSHIP

İÜC Career Club | Chair of the Board *Aug 2024 - Aug 2025*
- Built the annual plan of a 300+ member student club and coordinated 8+ events.

İÜC Machinery and Technologies Club | Member *Sep 2023 - Jul 2026*

Interests: 3D printing and prototyping | Woodcarving | Hiking